

RLPs, exchanges can accept and rank certain quotes and orders from certain participants in sub-penny increments as small as \$0.001.¹⁰⁵⁴ The national securities exchanges designed the RLPs to attract retail orders by providing a potential for price improvement at sub-penny levels because “most marketable retail order flow is executed in the OTC markets, pursuant to bilateral agreements, without ever reaching a public exchange” and OTC market makers typically pay retail brokers for their order flow.¹⁰⁵⁵ Quotes in RLP programs are not displayed. Instead, the appropriate SIP disseminates a flag indicating the side of the market for which an exchange has an RLP quote available at a price better than the NBBO. Because the exclusive SIP does not make known the price or the size of the RLP quote, market participants do not see the full liquidity available in RLP programs.¹⁰⁵⁶ To date, RLPs have not attracted a significant volume of retail order flow.¹⁰⁵⁷

One commenter stated the ability to trade in increments of smaller than a penny in certain circumstances as means of suggesting that there may be no need for rulemaking.¹⁰⁵⁸ However, it is not possible to post a displayed quote at an increment other than a penny. The economic forces

Program Approval Order); and Nasdaq BX Equity Rule 4780; Exchange Act Release No. 73702 (Nov. 28, 2014), 79 FR 72049 (Dec. 4, 2014) (NASDAQ BX Retail Pilot Program Approval Order).

¹⁰⁵⁴ See discussion in supra section III.C.1.

¹⁰⁵⁵ See specifically, NYSE Retail Liquidity Program Approval Order, supra note 1053, at 40679. The Commission stated that “[i]nternalizing broker-dealer[s] can offer sub-penny executions, provided that such executions do not result from impermissible sub-penny orders or quotations” by “typically select[ing] a sub-penny price for a trade without quoting at that exact amount or accepting orders from retail customers seeking that exact price.”

¹⁰⁵⁶ See, e.g., UTP Participant Input Specification (April 2024), available at https://www.utpplan.com/DOC/UtpBinaryInputSpec_Fractional.pdf.

¹⁰⁵⁷ See Proposing Release, supra note 11, at 80272 n.70 (citing to industry and academic discussions on why RLPs may not attract significant retail order flow).

¹⁰⁵⁸ See Lewis Letter at 33-34, attached to Virtu letter II.

that govern the tradeoffs in determining the tick size depend on the quote being displayed, ranked, and accepted. The empirical analysis also pertains to displayed quotes.

b. Analysis of Quoted Spreads under Current Ticks

This section discusses empirical analysis by the Commission on the prevalence of trading at different quoted spread ranges, defined in the Proposing Release using the concept of TWAQS.¹⁰⁵⁹ To document the prevalence of trading at different quoted spread ranges, Table 3 presents data on trading volume in 2023 based on average time weighted quoted spreads throughout the entire year for NMS stocks with a quotation, order, or indication of interest priced equal to or greater than \$1.00 per share.¹⁰⁶⁰ The analysis breaks trading volume each day into one of 17 average quoted spread bins, beginning with stocks that have TWAQS less than or equal to \$0.011.¹⁰⁶¹ Table 3 also reports the daily average number of stocks in each bin.

The data analysis in Table 3 indicates that under the current tick size regime in 2023, 65.2% of share trading volume (31.5% of dollar volume) occurred in stocks in the first average quoted spread bin of \$0.011 or less. Table 3 also reports that an additional 9.1% of share volume (15.3% of dollar volume) occurred in stocks with quoted spreads between \$0.011 and \$0.015. In

¹⁰⁵⁹ See Proposing Release, supra note 11, Table 4.

¹⁰⁶⁰ The data derived in Table 3 was derived using the same methodology as corresponding Table 4 in the Proposing Release (PR Table 4). See Proposing Release, supra note 11, at 80308. The only differences between the tables are that Table 3 uses data for all trading days in 2023 (instead of from Jan. to May 2022 in PR Table 4), and Table 3, because of our policy choice to set the minimum tick size at 0.5c (for quotes and orders priced \$1.00 or more for NMS stocks that have a TWAQS of \$0.015 or less), adds tick size bins for quoted spreads from 1.1c to 1.5c and 1.5c to 2c.

¹⁰⁶¹ Because of the \$0.01 minimum quoting increment for NMS stocks priced equal to or greater than \$1.00 per share, a stock cannot have a quoted spread less than \$0.01 unless markets become locked or crossed. The existence of locked and crossed markets can in some cases result in time weighted quoted spread that are very slightly lower than \$0.01. However, even for stocks with spreads most constrained by the tick, even a fraction of a second spent with a higher spread would likely result in an average quoted spread higher than \$0.01. For example, a large trade can exhaust liquidity deeper in the limit order book such that the stock's quoted spread temporarily increases from \$0.01. Thus, time weighed quoted spreads will virtually always be greater than \$0.01. This makes \$0.011 a more pragmatic minimum cutoff for empirical analysis than \$0.01.

sum, in Table 3, in 2023, approximately 74.3% of share volume (46.8% of dollar volume) transacted in NMS stocks (specifically, NMS stocks with a quotation, order, or indication of interest priced equal to or greater than \$1.00 per share) which have a quoted spread that is likely to be constrained by the minimum pricing increment,¹⁰⁶² which as discussed both above (section VII.B.2) and below (VII.D.1) increases transaction costs.

Table 3: Share Volume by Quoted Spread 2023^a			
Quoted Spread	Share Volume (%)	Dollar Volume (%)	Average # Stocks
Quoted Spread ≤ \$0.011	65.2%	31.5%	1,782
\$0.011 < Quoted Spread ≤ \$0.015	9.1%	15.3%	638
\$0.015 < Quoted Spread ≤ \$0.02	4.2%	5.2%	560
\$0.02 < Quoted Spread ≤ \$0.03	5.7%	8.4%	1,036
\$0.03 < Quoted Spread ≤ \$0.04	3.6%	7.6%	811
\$0.04 < Quoted Spread ≤ \$0.05	2.1%	3.6%	673
\$0.05 < Quoted Spread ≤ \$0.06	1.5%	2.3%	582
\$0.06 < Quoted Spread ≤ \$0.07	1.2%	2.1%	522
\$0.07 < Quoted Spread ≤ \$0.08	1.0%	1.8%	445
\$0.08 < Quoted Spread ≤ \$0.09	0.8%	1.6%	377
\$0.09 < Quoted Spread ≤ \$0.10	0.7%	1.6%	317
\$0.10 < Quoted Spread ≤ \$0.11	0.6%	1.5%	271
\$0.11 < Quoted Spread ≤ \$0.12	0.5%	1.3%	222
\$0.12 < Quoted Spread ≤ \$0.13	0.4%	1.1%	187
\$0.13 < Quoted Spread ≤ \$0.14	0.3%	1.1%	163
\$0.14 < Quoted Spread ≤ \$0.15	0.3%	1.0%	144
\$0.15 < Quoted Spread	2.8%	13.0%	2,177
^a This table provides share volume by stocks with different quoted spread profiles. To create this table, for each day the universe of stocks (identified by a unique stock variable) covered in the WRDS Intra-Day Indicators data are assigned into one of the 17 quoted spread bins based on that day's time-weighted quoted spread as computed by			

¹⁰⁶² For the share volume, 74.3% = 65.2% (i.e., Quoted Spread ≤ \$0.011) + 9.1% (i.e., \$0.015 < Quoted Spread ≤ \$0.02). For the dollar volume, 46.8% = 31.5% (i.e., Quoted Spread ≤ \$0.011) + 15.3% (i.e., \$0.015 < Quoted Spread ≤ \$0.02). See *supra* section VII.B.2 for a definition of tick-constrained; see also Proposing Release, *supra* note 11, Table 4, which used the same methodology and estimated that, in the first six months of 2022, 56% of share volume transacted in NMS stocks with a quoted spread of < \$0.011, while an additional 15% of share volume traded in stocks with a quoted spread of \$0.011 < and ≤ \$0.02.

WRDS Intra-Day Indicators. Then all share and dollar trading volume across all trading days in 2023 is aggregated for each of the 17 quoted spread bins. Percentages based on these totals are then computed. This table also presents the daily average number of stocks in each bin. To compute this variable, for each trading day in 2023, the number of stocks in each bin is tabulated, then the average across all trading days is presented here. Certain items in this Table 3 may also be affected by the MDI Rules once they are fully implemented. See infra section VII.C.3.

c. Reverse Stock Splits

A reverse split exchanges a fixed number of existing shares for a smaller number of new shares. The new shares have a higher price, but there are fewer of them.¹⁰⁶³ For example, if an issuer undergoes an 8-for-1 reverse split, eight shares are exchanged for one share that is worth the same as the eight, increasing the price by a factor of eight. All else equal, one would expect the quoted spread, which represents the per-share trading costs to also rise by a factor of eight. Thus, the cost of transacting in the stock, for a given dollar exposure, would remain constant. However, if a stock were tick constrained, undergoing a reverse split would cause the tick to become smaller relative to the (higher) stock price, thereby alleviating the tick constraint and reducing transaction costs.¹⁰⁶⁴

¹⁰⁶³ See, e.g., General Electric, GE Reverse Stock Split Frequently Asked Questions as of September 21, 2021, at 1, available at https://www.ge.com/sites/default/files/GE_Reverse_Stock_Split_FAQs.pdf (last accessed July 27, 2024). See also FINRA, Stock Splits, available at <https://www.finra.org/investors/investing/investment-products/stocks/stock-splits>.

¹⁰⁶⁴ For example, suppose a stock trades at \$10 per share and is tick-constrained such that it trades with a quoted spread of \$0.01, but could sustain a quoted spread of \$0.004 if not for the penny tick. If the stock undergoes a 5-for-1 reverse split, then the share price would rise to \$50 (5*\$10) and the quoted spread would rise to \$0.02 (5*\$0.004). The reverse split thereby alleviates the tick constraint. The reverse split also reduces trading costs in this case. To see the reduction in costs, suppose an investor wants to submit a market order for \$100 worth of the stock. With a \$10 price and a \$0.005 half-spread, this would entail purchasing five shares and paying a transaction cost of \$0.025 (i.e., five times the half-spread of \$0.005). With a \$50 price and a \$0.01 half-spread, however, the market order would only entail purchasing two shares and paying a transaction cost of \$0.02 (i.e., two times the half-spread of \$0.01), thus reducing transaction costs by 25%. See infra note 1295 and surrounding discussion for empirical studies documenting a reduction in trading costs when tick-constrained stocks complete a reverse split.

Using this logic, one commenter pointed out that currently the problem that the Commission identifies with regard to tick constrained stocks could be solved without Commission action by issuers making the decision to undergo a reverse stock split.¹⁰⁶⁵ The commenter further states, “[t]he fact that issuers do not discuss this indicates the issue is immaterial to them.”¹⁰⁶⁶ While the Commission agrees that a reverse split alleviates the tick constraint, there may be reasons that an issuer may choose not to undergo a reverse split apart from an assessment of the immateriality of tick constraints. Reverse splits have historically been associated with negative stock returns.¹⁰⁶⁷ Research suggests that this may be because the market views stock splits as a signal,¹⁰⁶⁸ and issuers may not wish to give the appearance of bad news with a reverse split. Reverse splits also change the share price, which may affect the trading characteristics of the stock.¹⁰⁶⁹ In addition, the direct administrative costs of a stock split for a large issuer are estimated to be between \$250,000 and \$800,000 (as of 2009).¹⁰⁷⁰ Finally, an

¹⁰⁶⁵ See Virtu Letter II at 25.

¹⁰⁶⁶ Id. at 25.

¹⁰⁶⁷ See, e.g., Hemang Desai & Prem C. Jain, Long-Run Common Stock Returns Following Stock Splits and Reverse Splits, 70 J. BUSINESS 3 (July 1997); Kim et al., Return Performance Surrounding Reverse Stock Splits: Can Investors Profit?, 37 FIN. MGMT. 2 (Summer 2008).

¹⁰⁶⁸ For a discussion on the signal conveyed by stock splits, see, e.g., Maureen McNichols & Ajay Dravid, Stock Dividends, Stock Splits, and Signaling, 45 J. FIN. 3 (July 1990); David L. Ikenberry et al., What Do Stock Splits Really Signal?, 31 J. FIN. & QUANTITATIVE ANALYSIS 3 (Sept. 1996). The academic literature on signaling generally uses stock splits (rather than reverse splits) because standard databases do not include the announcement date of reverse splits, which makes the measurement of the market reaction to the announcement of a reverse split imprecise. Once the reverse split is completed, however, the academic literature documents negative returns—see id.

¹⁰⁶⁹ See, e.g., Kelly Shue & Robert Townsend, Can the Market Multiply and Divide? Non-Proportional Thinking in Financial Markets, 76 J. FIN 5 (Oct. 2021). This paper documents greater return responses to news in lower-priced stocks and hypothesizes that some investors think about stock price changes in terms of dollars rather than returns. The paper studies reverse stock splits and finds evidence consistent with this hypothesis.

¹⁰⁷⁰ See Weld et al., The Nominal Share Price Puzzle, 23 J. ECON. PERSPECTIVES 2 (Spring 2009). The authors estimate the administrative costs of stock splits. Reverse stock splits are likely to have similar administrative costs. After adjusting for changes in the consumer price index from 2009 to 2024, the paper’s estimated direct administrative cost of a split for a large issuer in 2024 is \$365,000 to \$1,170,000.

issuer may not capture all of the benefits of the liquidity arising from the reverse split and the alleviated constraints. In sum, though reverse stock splits may address tick constraints, they are an inefficient solution for tick constrained securities. One commenter agreed, stating that “while issuers can make pricing in their securities more or less granular through reverse or forward splits, it’s not practical for the SEC and the industry to rely on them to do so.”¹⁰⁷¹

2. Access Fees

This section discusses current regulation on the access fee cap, the current practices at exchanges for setting access fees and rebates, and Commission analysis showing that most exchanges assess access fees close to the current access fee cap and use these access fees to principally fund rebates.

a. Current Regulations

Preexisting Rule 610(c) limits the fees that trading centers can charge for accessing protected quotations with prices of \$1.00 per share or greater to \$0.0030 per share (or 30 cents per 100 shares). This level is commonly referred to as 30 mils.¹⁰⁷² Preexisting Rule 610 also prohibits access fees in excess of 0.3% of the price for stocks priced less than \$1.00 per share. The 30 mil fee cap was adopted as a part of Regulation NMS in conjunction with the order protection rule and was implemented to prevent trading centers from charging excessive fees to orders that were required to trade with a protected quote.¹⁰⁷³ The 30 mil fee cap was also set

¹⁰⁷¹ See The Tick Size Debate Revisited attached to MEMX Letter at 69.

¹⁰⁷² See Proposing Release, supra note 11, at 80309.

¹⁰⁷³ See Regulation NMS Adopting Release, supra note 4, at 37545 (justifying the 30 mil limit: “For quotations to be fair and useful, there must be some limit on the extent to which the true price for those who access quotations can vary from the displayed price To protect limit orders, orders must be routed to those markets displaying the best-priced quotations. This purpose would be thwarted if market participants were allowed to charge exorbitant fees that distort quoted prices”); see also supra note 434 and surrounding discussion from the Regulation NMS Adopting Release on the potential—absent a fee cap—for high fee markets to take advantage of intermarket price protections.

based on existing market practices at the time.¹⁰⁷⁴ Rule 610(c) only regulates fees to access protected quotes; it does not regulate fees to access non-protected quotes, nor does it regulate rebates that exchanges can offer. However, the 30 mil fee cap has become a central component of the structure of fees and rebates as access fees for non-protected quotes generally do not exceed the 30 mil fee cap, nor do typical rebates.¹⁰⁷⁵

b. Current Practices at Exchanges

The transaction fee structure on an exchange currently takes one of three forms. The most common is maker-taker, in which liquidity demanders (i.e., takers) are assessed the access fee and liquidity providers (i.e., makers) are offered a rebate. Exchanges can also be inverted (also known as taker-maker), in which liquidity demanders are offered a rebate and liquidity providers are assessed an access fee.¹⁰⁷⁶ The last form of fee structure is flat; a flat exchange either charges one or both sides a fee but does not offer rebates. While the exchanges are free to subsidize rebates beyond what they earn through collecting access fees, in practice this does not appear to happen.¹⁰⁷⁷ The difference between the average access fee charged and the average rebate paid is the net capture earned by the exchanges for facilitating a transaction.¹⁰⁷⁸

¹⁰⁷⁴ See id. at 37503 (the 30 mil access fee cap was chosen because “it will not seriously interfere with current business practices” and “[i]n the absence of a fee limitation, some ‘outlier’ trading centers might take advantage of the requirement to protect displayed quotations by charging exorbitant fees to those required to access the outlier’s quotations”); see also supra note 357.

¹⁰⁷⁵ See infra Table 4 for a summary of transaction-based fee schedules for U.S. national equities exchanges as of Feb. 2024.

¹⁰⁷⁶ While Rule 611 creates incentives for exchanges to use high fees and rebates in order to quote at the NBBO—see infra note 1120—not all exchanges compete on this margin (e.g., inverted exchanges).

¹⁰⁷⁷ See infra section VII.D.2 for more discussion on why exchanges may not subsidize rebates from other sources of revenue; see also Eric Budish, et al., A Theory of Stock Exchange Competition and Innovation: Will the Market Fix the Market? (working paper May 22, 2019) available at <https://ssrn.com/abstract=3391008> (retrieved from SSRN Elsevier database).

¹⁰⁷⁸ See Proposing Release, supra note 11, at 80291 n.304 (“Net capture” is the amount earned by the trading center for facilitating a transaction, which is typically the difference between the average access fee

The regulatory access fee cap is most relevant for maker-taker markets where the trader accessing a protected quote must pay the access fee. This is because the access fee cap applies only to fees for accessing protected quotations and does not apply to fees for posting quotations. On an inverted venue, the exchange is not restricted by preexisting Rule 610 in terms of the rebate that it can offer to access a protected quote or the fee to post a protected quote.¹⁰⁷⁹ Flat rate venues, which do not offer rebates, do not appear to be economically constrained by the preexisting Rule 610(c) as their fees for both taking and adding liquidity are significantly lower than the 30 mil fee cap.¹⁰⁸⁰

Fee/rebate schedules can be quite complex, and the fee schedules change frequently.¹⁰⁸¹ As was discussed in the Proposing Release,¹⁰⁸² the actual fee or rebate that an exchange member is assessed on most exchanges also generally depends on which tier a market participant falls into based on trading volume in that month, with higher-volume market participants typically receiving a higher rebate or a lower fee.¹⁰⁸³ Exchanges file their fee and rebate schedules with

charged by the trading center and the average rebate paid by the trading center”). It is common practice across exchanges to fund their rebates with transaction fees. In principle rebates could exceed access fees as, unlike access fees, there is no regulatory cap restricting the rebates that can be offered. However, it is unlikely that trading venues would offer rebates in excess of the fees collected as doing so would expose them to the possibility of large losses. See supra note 1101 and accompanying text for further discussion on exchanges’ net capture rates.

¹⁰⁷⁹ As can be seen from Table 4, which presents information on access fees and rebates for the 16 operating exchanges, in practice the fee that is charged on an inverted fee venue to post liquidity is generally very close to the 30 mil access fee cap even though not constrained by Rule 610.

¹⁰⁸⁰ As can be seen from Table 4, the only flat exchange (LTSE) is one that does not levy either a fee or rebate and prior to adopting a maker-taking fee model another exchange (IEX) had charged both sides of the transaction 9 mils. See Proposing Release, supra note 11, at 80311.

¹⁰⁸¹ See Table 4 for information on how often exchanges amend their fees.

¹⁰⁸² See Proposing Release, supra note 11, at 80292.

¹⁰⁸³ See Letter from Richard Steiner, Electronic Trading Strategist, RBC Capital Markets, to Brent Fields, Secretary, Commission (Oct. 16, 2018), available at <https://www.sec.gov/comments/s7-05-18/s70518-4527261-176048.pdf> (commenting on the transaction fee pilot); see also the Fee Tiers Proposal, supra note 437.

the Commission and post them on their websites. While this means that the rebate and fee rates associated with each volume-based tier can be known at the time a market participant trades, market participants may not know which volume-based tier they will fall under at the time of the trade (and thus the fee or rebate rate that will apply to their particular trade) because the tier they will fall under is typically determined based on their trading volume during the current month, which is not finalized until the end of the month.¹⁰⁸⁴ More specifically, the volume-based fees or rebates a market participant receives from an exchange are often determined by a market participant's average total daily traded share volume on the exchange during the month as a percentage of either the average total daily market volume reported by one of the consolidated tapes during the month or as a percentage of the average total daily market volume reported by all consolidated tapes during the month.¹⁰⁸⁵ It is therefore, as one commenter stated, difficult for

¹⁰⁸⁴ See Chester Spatt, Is Equity Market Exchange Structure Anti-Competitive? (Dec. 28, 2020), available at <https://www.cmu.edu/tepper/faculty-and-research/assets/docs/anti-competitive-rebates.pdf>. However, not all exchanges offer volume-based tiers in their fee structures. For example, LTSE does not charge fees to transact. For exchanges like these, it is possible to determine with certainty the cost to transact prior to executing a trade.

¹⁰⁸⁵ The Equity Data Plans disseminate SIP data over three separate networks: (1) Tape A for securities listed on the New York Stock Exchange ("NYSE"); (2) Tape B for securities listed on exchanges other than NYSE and Nasdaq; and (3) Tape C for securities listed on Nasdaq. These tapes are referred to as the "consolidated tapes." The CTA Plan governs the collection, consolidation, processing, and dissemination of last sale information for Tape A and Tape B securities. The CQ Plan governs the collection, consolidation, processing, and dissemination of quotation information for Tape A and Tape B securities. Finally, the UTP Plan governs the collection, consolidation, processing, and dissemination of last sale and quotation information for Tape C securities. For details on exchange volume based fees and rebates, see, e.g., Add and Remove Rates, Nasdaq, available at <http://www.nasdaqtrader.com/Trader.aspx?id=PriceListTrading2>; New York Stock Exchange Price List 2024, NYSE, available at https://www.nyse.com/publicdocs/nyse/markets/nyse/NYSE_Price_List.pdf; and Cboe U.S. Equities Fee Schedules EDGX Equities, CBOE, available at https://www.cboe.com/us/equities/membership/fee_schedule/edgx/. See also the Fee Tiers Proposal supra note 437 at 76284-88, describing Commission concerns about the effect of tiers on competition among exchange members, conflicts of interest between members and their customers, and competition between exchanges.

market participants to forecast trading costs.¹⁰⁸⁶ Hence, market participants currently typically have to make trading decisions without the ability to determine their full trading costs.

Broker-dealers trading in an agency capacity may pass fees and rebates received from exchanges while working a customer's order through to the customer, either directly or through a change in the commission charged for the order. One commenter stated that passing through transaction fees and rebates is not common.¹⁰⁸⁷ The Commission is uncertain the extent to which transaction fees and rebates are passed through to customers, or in what form. It is possible that fees and rebates are passed through indirectly in the form of payment for order flow or in price improvement. For example, one broker-dealer's 606 reports, in discussing orders routed to a wholesaler, acknowledged that exchange rebates may affect the wholesaler's subsequent routing decision, but that those rebates could be used to provide price improvement to the broker-dealer's customers or order flow payments to the broker-dealer.¹⁰⁸⁸ As discussed in the Proposing Release, the Commission is uncertain of how much demand currently exists for rebates to be passed through to end customers.¹⁰⁸⁹

Market participants who are not themselves broker-dealers may access information on exchange fees and rebates through reports available under Rule 606. With respect to held orders,

¹⁰⁸⁶ See Council of Institutional Investors Letter at 4 (stating "It is our understanding that currently exchanges use volume-based tier schedules that depend on the current month's trading volume. As a result, the per-transaction fee or rebate cannot be known when the trade occurs. This significantly impedes the ability of institutional investors and other market participants 'to evaluate the total price of a trade at the time of execution and . . . [the] ability to evaluate best execution and order routing.'").

¹⁰⁸⁷ See Harris Letter at 12 ("Almost all retail and many institutional brokers pay the taker fee and keep the maker rebate when trading on behalf of their clients.").

¹⁰⁸⁸ See E*Trade.com, Morgan Stanley Smith Barney LLC - Held NMS Stocks and Options Order Routing Public Report, 1st Quarter, 2024, at 2, [available at https://cdn2.etrade.net/1/24043013500.0/aempros/content/dam/etrade/retail/en_US/documents/pdf/order-routing-reports/2024/606-MSWM-2024Q1.pdf](https://cdn2.etrade.net/1/24043013500.0/aempros/content/dam/etrade/retail/en_US/documents/pdf/order-routing-reports/2024/606-MSWM-2024Q1.pdf).

¹⁰⁸⁹ See Proposing Release, *supra* note 11, at 80330.